

2

Calibration

2.1 Direct and Inverse Calibration

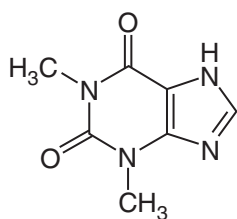
Analytical methods are based on well-known physicochemical or biochemical phenomena which can be described by various equations. To illustrate the practical problems when calibrating a method of analysis, we can refer to techniques widely used in laboratories. In conventional ultraviolet–visible (UV–Vis) spectrophotometry, the ratio of the emitted flux Φ_0 to the transmitted flux Φ after passing through the solution being measured, or transmittance, is used for instrumental response. Depending on the type of spectroscopic method, the logarithm of the transmittance is called the absorbance, transmittance, or optical density (*OD*). According to Beer's law, this ratio depends on the concentration X of the solution, the length L of the measuring cell, and a molar extinction coefficient ε_T which depends on the temperature T and, of course, the nature of the analyte giving the well-known Beer's law theoretical model:

$$\log\left(\frac{\Phi_0}{\Phi}\right) = \varepsilon_T \times X \times L$$

In all cases, the logarithmic transformation of the flux ratio aims to linearize the response function. As explained in Section 1.1.3, this pretreatment is universal and invisible to the analyst. Obviously, it would be sufficient to know the value of ε_T to predict the responses depending on the values of X . Unfortunately, this coefficient varies greatly when the emitted radiation is not purely monochromatic or when the temperature is not constant. In practice, Beer's law cannot be directly used for calibration, since the measuring system is not isolated and is subject to random disturbances due to environmental variations, such as temperature, atmospheric pressure, instability of the electrical voltage, sound vibrations, aging of the equipment components, etc. All these disturbances together form the background noise, which can be estimated, measured, and subtracted.

This simple example shows how calibration must rely on a more complex model than initially imagined from the fundamental equations of physics or chemistry. To illustrate the statistical issues of calibration, we are taking a method of analysis more complex than spectrophotometry, involving mass spectrometry. This chapter is limited to external calibration as it represents the most commonly used calibration

mode. Concerning internal calibration or in-sample calibration, several modeling approaches were already presented in Chapter 1.



Theophylline

The same example will be used in different chapters as a roadmap to illustrate the proposed strategy for method validation and measurement uncertainty (MU) estimation. This is a method of analysis of theophylline in human plasma. More details about the complete THEOPHYLLINE method and dataset are given in Table 5.1 and in Section 5.2.2, introducing the method accuracy profile (MAP) procedure.

Theophylline is a molecule used to manage the symptoms of asthma and other lung ill-conditions caused by reversible airflow obstruction. It works by relaxing the smooth muscles in the bronchial airways and pulmonary blood vessels. The aim of the study used for this example was to develop and validate a method for the quantitative determination of different xanthines, including theophylline, caffeine, theobromine, and paraxanthine, in human plasma. This was achieved within the framework of a sports medicine project interested in the effect of these molecules on athletes' performance. These analytes are determined by ultra-high performance liquid chromatography (UHPLC), coupled with a tandem mass spectrometer detection UHPLC-MS/MS [1].

Five calibrators were prepared and contained the analytical grade theophylline (99% purity) at 0.02, 0.1, 0.5, 2.5, and 10.0 $\mu\text{g/l}$. For each calibrator, two replicates were prepared. As explained in Section 5.2.2, this calibration experimental design was repeated over six different days giving six series of similar calibration datasets. Table 2.1 presents only one of these series or days, called series 1. Other series are comparable and give the same conclusions. The graphical illustration of these data is straightforward when using a worksheet.

The observed points seem to be correctly fitted to a straight line. With this software, it is easy to quickly add on the same graphics several trendlines, such as straight-line, second-order, or quadratic polynomial. When looking at the r^2 values reported in

Table 2.1 THEOPHYLLINE – raw calibration of series 1 dataset, measurement values are expressed in arbitrary units (AU).

Concentration ($\mu\text{g/l}$)	Replicate 1	Replicate 2
0.02	0.293	0.443
0.10	1.874	1.810
0.50	8.904	8.306
2.50	23.411	37.832
10.00	124.84	129.605

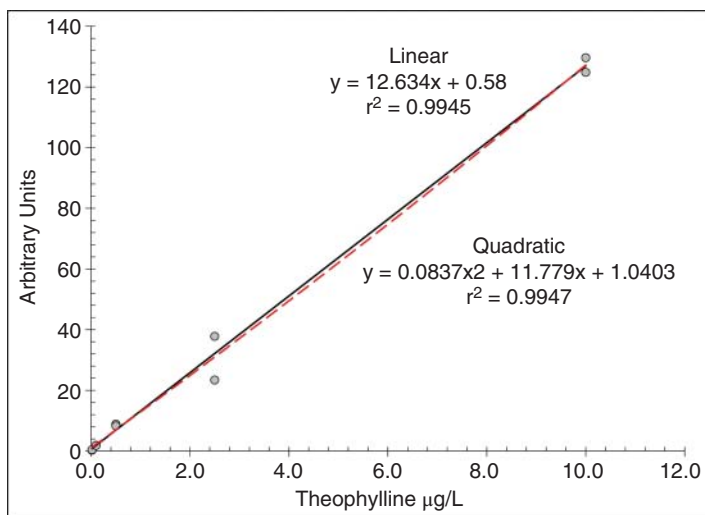


Figure 2.1 THEOPHYLLINE – illustration of the calibration data of series 1. Solid line: linear model. Dashed line: quadratic model.

Figure 2.1, both models seem equivalent and well-matched. But this simple visual evaluation is misleading, as shown further in Section 8.1. The general statistical model that is used to establish the calibration curve can be described by Eq. (2.1):

Generic direct calibration curve

$$Y = f(X_c) \quad (2.1)$$

- X_c the known concentration of the calibration compound (authentic or surrogate, as explained in Section 1.1.2).
- Y the measured instrumental response.
- f the deterministic model chosen for the response function.

As a reminder, the aim of calibration is not to prove that some relationship between the concentration and the instrumental response exists. All quantitative analysis methods are founded on well-known physicochemical or biological mechanisms, and it is well established that such a relationship exists. In that context, calibration is a two-step procedure:

- Step 1: *Direct calibration*. It consists of collecting instrumental responses of calibrators and calculating the calibration curve coefficients that most accurately relate the (known) calibrant concentration to the instrumental response.
- Step 2: *Inverse calibration*. It uses the inverse function of the calibration model to predict the concentration of unknown samples based on the instrumental response they provide. The values obtained by this operation are called *inverse-predicted* concentrations. Sometimes they can also be called back-calculated, but this may be confusing as the back-calculation procedure is employed in various fields of technology, such as real-time process monitoring, to restore previous situations from historical data. The mathematical interpretation

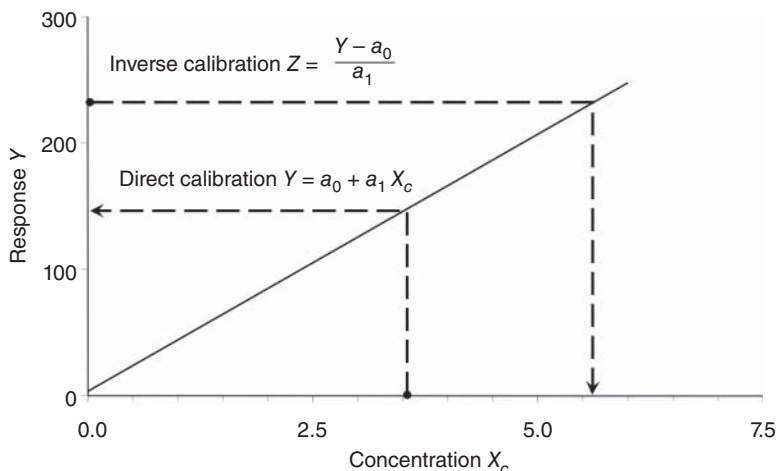


Figure 2.2 Direct calibration and inverse calibration.

of this step is summarized by Eq. (2.2), which gives its name to the inverse calibration:

Inverse calibration function

$$Z = f^{-1}(Y) \quad (2.2)$$

The two-step quantification procedure is illustrated in Figure 2.2 in the case of a linear calibration model. With theophylline series 1 calibration data, two models, graphically close, can be adjusted, namely linear and quadratic. The question of selecting the best model is raised.

A viable way to answer is to collect new data from samples with known contents, estimate their inverse predicted concentrations and verify which model gives the best inverse predictions. It remains to define what means *the best* calibration model. The role of inverse calibration is quite specific to analytical sciences, and several authors proposed to organize calibration in one stage to avoid this inversion of the calibration curve. The calibration would be named *reverse* calibration as discussed in Section 2.5.

2.2 Least-squares Regression Method

2.2.1 Straight-line Computation

In the case of multipoint (external) calibration, it is necessary to prepare several calibrators and estimate the calibration model. For simplicity, let us denote X_i the known calibrant concentration of the i th calibrator. To fully establish the statistical model of the calibration curve, it is necessary to include the background noise noted E in Eq. (2.1). As stated before, it results from various sources of disturbances, such as

instrument instability or calibrator preparation. If E_i is the random variable accounting for the background noise of the instrumental response Y_i measured on calibrator X_i , the observed model is given by Eq. (2.3).

Number of calibration measurements

$$1 \leq i \leq I$$

Measured instrumental response

$$Y_i = f(X_i) + E_i \quad (2.3)$$

Predicted instrumental response

$$\hat{Y}_i = f(X_i) \quad (2.4)$$

Residual

$$E_i = Y_i - \hat{Y}_i \quad (2.5)$$

A part of the model is deterministic and corresponds to $f(X)$ the other is random. The determinism is related to the physicochemical or biological phenomena used to explain that Y is modified when X varies. To make clear this dual structure of the model, statisticians introduced the notation $\hat{Y}_i$ (hat) corresponding to a predicted value of Y once f is known. The difference between measured and predicted instrumental responses corresponds to the random part, i.e. unpredictable, of the model and is called the residual. It is defined by Eq. (2.5).

In Figure 2.3 the linear model is used to illustrate these concepts and explain how the least-squares algorithm works for estimating the coefficients of the f function but it can be transposed to any other model. The observed linear model is given by Eq. (2.6). In this case, the intercept a_0 is interpreted as the blank, and the slope a_1 as the sensitivity of the method. Knowledge of these coefficients is essential to perform the inverse calibration stage.

Observed linear calibration curve

$$Y_i = a_0 + a_1 X_i + E_i \quad (2.6)$$

Theoretical linear calibration curve

$$\hat{Y}_i = a_0 + a_1 X_i \quad (2.7)$$

The ordinary least-squares (OLS) method is the most widespread algorithm for estimating a_0 and a_1 . A major assumption so that OLS can satisfactorily apply is that the two variables Y and X must play different roles. Y is explained or dependent, while X is said to be explanatory or independent. This means there is a causal relationship between the two variables. Because the instrumental response Y is fully explained by the variations of the concentration X , calibration is a typical example for least-squares application. This remark impacts the nature of the residual random variable E_i . In practice, it is realistic to suppose it is formed of two components: E_Y linked to the instrumental background and E_X linked to the errors of preparation of the calibration solutions.

$$E_i = E_X + E_Y$$